

Real-Time Media Intelligence & Narrative Risk Platform

Project Report

Executive Summary

This project is an end-to-end streaming data platform that continuously ingests news from 15+ global sources, applies AI-powered enrichment across five intelligence dimensions, and delivers actionable insights through a live dashboard and REST API. The platform processes 200+ articles per hour, covers 13 countries, tracks 40+ news sources with individual credibility scores, and detects narrative anomalies using statistical Z-score analysis — all running on open-source tooling with zero licensing cost.

The system is architected around the principle of decoupled streaming: every component is independently scalable and replaceable. The ingestion layer does not know about enrichment. The enrichment layer does not know about storage. The serving layer does not know about ingestion. This separation mirrors how production data platforms at companies like Bloomberg, Meltwater, and Brandwatch are structured.

Problem Statement

Organizations monitoring media — brand managers, political analysts, investment teams, policy researchers — currently have two options: hire human analysts to read news manually, or pay ₹50,000–₹2,00,000 per month for commercial tools like Meltwater, Brandwatch, or Dow Jones Factiva. Neither option scales well. Manual monitoring misses signals. Commercial tools are opaque, expensive, and inflexible.

The core problems this platform solves:

- **Signal latency:** By the time a crisis appears on executive radar, it has already trended for hours. The platform detects anomalies 10-15 minutes after emergence.
- **Source inequality:** A Reuters article and a random blog carry the same weight in simple keyword monitoring. The platform applies credibility-weighted sentiment so tier-1 sources drive the signal more than low-credibility noise.
- **Keyword brittleness:** Predefined keyword lists miss synonyms, paraphrasing, and emerging terminology. The platform uses semantic embeddings to cluster narratives without any predefined rules.
- **Geographic blindness:** Most tools give global sentiment scores. The platform provides country-level sentiment breakdown, letting teams identify regional crises

before they become global ones.

Architecture Overview

The platform is organized into five layers, each with a single responsibility.

Layer 1 — Ingestion

Three data sources feed the pipeline:

RSS Ingestor: Polls 15+ RSS feeds every 3 minutes using feedparser. Sources include The Hindu, Mint, Economic Times, Business Standard, Reuters, BBC, NYT, TechCrunch, and others. Each article is deduplicated using an in-memory URL set within a session. The ingestor extracts title, summary, full text where available, published timestamp, and source URL.

NewsAPI Ingestor: Queries the NewsAPI free tier on 7 predefined topic queries — data breaches, CEO scandals, AI innovation, layoffs, election fraud, product recalls, and IPO/funding news. Runs every 15 minutes to respect free tier rate limits. Returns structured article metadata with publication timestamps.

Source Registry: Every article is enriched at ingestion time with a credibility score from a hand-curated registry of 40+ outlets. Scores range from 0.95 (Reuters, AP) to 0.35 (known low-credibility outlets). The registry uses domain extraction from URLs and falls back to 0.50 for unknown sources. This score flows through every downstream calculation.

All three ingestors publish a canonical `RawArticle` event — a dataclass with 14 fields — to Redpanda using confluent-kafka.

Layer 2 — Message Broker

Redpanda serves as the central message bus. It implements the Kafka protocol in a single C++ binary — no ZooKeeper, no JVM, dramatically lower resource consumption than Apache Kafka while maintaining full API compatibility.

Two topics are maintained:

- `raw-media-events` — immutable record of every ingested article, 3 partitions, 7-day retention, partitioned by source domain
- `enriched-media` — enriched events after ML processing, 3 partitions, 7-day retention

The separation of raw and enriched topics enables historical replay: if a better sentiment model is deployed, raw events from the past 7 days can be replayed through the new enrichment job without re-ingesting anything.

Layer 3 — Enrichment Engine

A Python consumer (replacing Spark Structured Streaming for single-node deployment) reads from `(raw-media-events)` and applies five enrichment steps to every article:

FinBERT Sentiment Scoring: ProsusAI/finbert, a BERT model fine-tuned on financial news corpus. Returns positive/negative/neutral label with confidence score. Superior to general-purpose sentiment models (VADER, TextBlob) for news content because it understands financial language nuances — "earnings missed by less than expected" scores differently from "earnings missed."

Narrative Clustering: all-MiniLM-L6-v2 sentence transformer embeds article title + summary into a 384-dimensional vector. New articles are compared against all existing cluster centroids using cosine similarity. If similarity exceeds 0.72, the article joins the closest cluster and updates its centroid via running average. Below threshold, a new cluster is created. This discovers themes like "Tamil Nadu elections", "OPEC oil shifts", "stagflation risk" entirely from article content — no predefined categories.

Entity Extraction: spaCy `en_core_web_sm` NER model extracts organizations, people, and locations from article text. Results stored as arrays in ClickHouse for arbitrary filtering. Powers the competitor comparison feature.

Geographic Inference: Location entities are mapped to ISO 3166-1 alpha-2 country codes using a 60-entry location dictionary covering major countries, Indian states and cities, and major global cities. Falls back to source region from the registry if no location entities are found.

Influence Weighting: $\text{Weighted sentiment} = \text{raw sentiment score} \times \text{source credibility}$, with sign applied based on label (positive = +, negative = −, neutral = 0). This produces a signed float where a highly credible negative source produces a strong negative signal and a low-credibility negative source produces a weak one. One Reuters negative article (0.95 credibility) outweighs approximately 19 low-credibility negative articles (0.50 credibility).

The enriched event — 29 fields total — is published to `(enriched-media)`.

Layer 4 — Storage & Transformation

ClickHouse receives enriched events via a batch sink consumer. Events are inserted in batches of 50 for efficiency. The main `(enriched_articles)` table uses MergeTree engine, partitioned by month on `(event_date)`, ordered by `((enriched_at, source_domain, sentiment_label))` for fast time-series aggregations.

A materialized view (`(mv_narrative_hourly)`) continuously computes per-hour narrative statistics as events land, maintaining a pre-aggregated `(narrative_hourly_stats)` table with a 30-day TTL.

dbt Core runs five transformation models on top of ClickHouse:

Model	Powers	Key computation
<code>mart_sentiment_by_source</code>	Source influence dashboard	Positivity % per source per hour
<code>mart_narrative_clusters</code>	Trending topics panel	Mention count + weighted sentiment per cluster per hour
<code>mart_geo_sentiment</code>	Geographic heatmap	Country-level avg sentiment per day
<code>mart_competitor_tracker</code>	Entity comparison	Controversy score = negative% per entity per hour
<code>mart_crisis_signals</code>	Crisis detection	Z-score on rolling 7-day baseline per narrative

dbt runs every 30 minutes via a Python scheduler, keeping mart tables fresh for dashboard queries.

Layer 5 — Serving

Grafana connects to ClickHouse via the official ClickHouse datasource plugin. The dashboard exposes four panels with a global time-range variable (`time_filter`) controlling all panels simultaneously:

- Sentiment Distribution — pie chart of positive/negative/neutral split
- Top Entities by Mention — bar chart ranked by weighted mention count
- Geographic Sentiment — country-level sentiment bar chart
- Trending Topics — table with topic, mention count, sentiment, tone indicator
- Crisis Signals — table of anomalous narratives with crisis probability

FastAPI exposes 8 REST endpoints for programmatic access. Anonymous viewer mode is enabled in Grafana for public dashboard access without login.

Intelligence Features

Narrative Detection

Articles cluster into themes automatically using semantic similarity. No keyword lists. No predefined categories. The system discovered "West Bengal Assembly polls", "OPEC oil

production shifts", "AI adoption", and "stagflation dangers" from raw article text within the first ingestion session. New narratives emerge within minutes of first coverage.

Source Influence Weighting

All 40+ tracked sources carry individual credibility scores informed by editorial standards, fact-checking processes, ownership transparency, and historical accuracy track records. Scores broadly align with independent media credibility organizations (NewsGuard, MBFC, Ad Fontes Media). The weighted sentiment calculation ensures signal quality — high-credibility coverage moves the needle, low-credibility noise is dampened.

Geographic Sentiment Heatmap

From the first 968-article dataset: India -0.15 (mildly negative, election coverage mixed with economic news), US -0.10 (AI/tech positivity offsetting political negativity), Brazil -0.80 (economic and political crisis coverage), Israel -0.65 and Ukraine -0.65 (war coverage). A global brand manager can immediately see which markets have deteriorating media perception.

Competitor Comparison

Any two entities appearing in extracted organizations can be compared. From live data: BJP 21 mentions, -0.378 sentiment, 57% controversy. Congress 10 mentions, -0.453 sentiment, 70% controversy. AI 7 mentions, $+0.052$ sentiment, 28% controversy — the only entity with positive sentiment in the dataset, reflecting optimistic technology coverage. The controversy score (negative mention percentage) is independently useful for risk assessment.

Early Crisis Detection

The Z-score model computes per-narrative baseline statistics over a rolling 7-day window. When a narrative's hourly article count exceeds 2 standard deviations above baseline AND weighted sentiment drops below -0.3 , crisis probability fires as HIGH. Thresholds are configurable in a single dbt SQL file. The model requires minimum 2 weeks of continuous ingestion to establish a reliable baseline — standard for anomaly detection systems.

Technical Decisions & Rationale

Redpanda over Apache Kafka: Single binary, no ZooKeeper dependency, 5× lower memory consumption. Full Kafka protocol compatibility means zero code changes if migrating to full Kafka in production.

ClickHouse over PostgreSQL: Columnar storage enables sub-second aggregations across millions of rows. "Average sentiment by country for the last 30 days" across 1M articles runs

in under 1 second on ClickHouse vs minutes on PostgreSQL. Essential for a real-time dashboard.

FinBERT over VADER/TextBlob: Domain-specific training on financial news corpus. Understands context that general models miss. "The stock fell less than expected" scores positive on FinBERT (beats expectations), negative on VADER (the word "fell").

Sentence-transformers over keyword matching: Keywords break on synonyms and paraphrasing. Embeddings capture semantic meaning. "RBI rate hike", "Reserve Bank holds repo rate", "India central bank decision" cluster together correctly with cosine similarity — they share no exact keywords.

dbt over raw SQL scripts: Business logic (crisis thresholds, controversy score formula, Z-score window) lives in version-controlled, documented, testable SQL. Changing the crisis threshold from $Z > 2.0$ to $Z > 1.5$ is a one-line change in one file. Without dbt it would require finding and updating multiple queries across the codebase.

Confluent-kafka over kafka-python: kafka-python has unresolved issues with Redpanda's broker metadata advertisement in Docker networking environments. confluent-kafka (librdkafka binding) handles broker address family resolution correctly with the `broker.address.family: v4` configuration option.

Dataset & Results

Metric	Value
Total articles processed	968
Ingestion sources	15+ RSS feeds + NewsAPI
Countries detected	13
Narrative clusters formed	~470
Unique entities tracked	50+
Sentiment distribution	45% neutral, 36% negative, 19% positive
Top entity by mentions	BJP (21 mentions, -0.378 weighted sentiment)
Most negative country	Brazil (-0.80 avg sentiment)
Pipeline latency	< 60 seconds publish to dashboard
ClickHouse insert throughput	968 rows in < 10 seconds

Metric	Value
dbt model build time	6 models in 4.39 seconds

Project Structure

```

news-sentiment-pipeline/
├── ingestion/
│   ├── models/article.py          # Canonical 14-field event schema
│   ├── sources/registry.py        # 40+ source credibility scores
│   ├── sources/rss_ingestor.py     # RSS feed producer
│   ├── sources/newsapi_ingestor.py # NewsAPI producer
│   └── utils/producer.py          # Shared Kafka producer
├── processing/
│   ├── models/sentiment.py         # FinBERT pipeline
│   ├── models/narrative.py         # Embedding + clustering
│   ├── models/entities.py         # spaCy NER
│   ├── utils/geo.py               # Location → country mapping
│   └── enrichment_job.py           # Main consumer + enrichment
├── storage/
│   ├── clickhouse/schema.sql       # Table + materialized view DDL
│   ├── clickhouse/sink.py          # Batch ClickHouse writer
│   └── dbt/models/
│       ├── staging/                # Raw → clean view
│       └── marts/                  # 5 intelligence feature tables
├── serving/
│   └── api/
│       ├── main.py                 # FastAPI app + CORS
│       ├── db.py                   # ClickHouse client
│       └── routers/                # 5 feature routers
├── docker-compose.yml              # Local dev stack
├── run_pipeline.sh                 # Single-command startup
├── run_dbt_scheduler.py            # 30-min dbt refresh
└── requirements.txt

```

Tech Stack Summary

Category	Technology	Version
Language	Python	3.11
Message broker	Redpanda	Latest
Stream processing	confluent-kafka	Latest
Sentiment model	FinBERT (ProsusAI)	HuggingFace
Embedding model	all-MiniLM-L6-v2	HuggingFace
NER model	spaCy en_core_web_sm	3.x
OLAP database	ClickHouse	Latest
Data transformation	dbt Core + dbt-clickhouse	1.8.0
Cache	Redis	7-alpine
API framework	FastAPI + Uvicorn	Latest
Dashboard	Grafana	Latest
Containerization	Docker + Docker Compose	Latest
Infrastructure	Oracle Cloud (E2.1.Micro)	—

Limitations & Future Roadmap

Current limitations:

- Crisis detection requires 2+ weeks of continuous ingestion to establish reliable Z-score baselines
- Entity deduplication is imperfect — "BJP", "Bharatiya Janata Party", and "ruling party" may appear as separate entities
- Narrative cluster labels use article titles rather than generated theme names
- Geographic inference relies on entity extraction accuracy — articles without named locations fall back to source region

Planned improvements:

- Twitter/X API integration for social sentiment layer
- LLM-generated narrative labels (GPT-4/Claude) replacing raw title labels
- Slack webhook alerts for HIGH crisis probability events
- Historical trend comparison (this week vs same week last month)
- Entity resolution and deduplication using fuzzy matching
- Multilingual support for Hindi and Tamil news sources
- Automated A/B testing of sentiment model versions using stored raw events

Built as a portfolio demonstration of production data engineering and GenAI integration patterns. All tooling is open-source. Infrastructure cost: \$0 (Oracle Cloud Always Free tier).